

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-aa61f357b764ba56e.jsonl`  
- SHA-256: `072fde9e1409e0a8068cb9fc37067900b141a95febc78577b907497b78dd56eb`  
- Source modified: `2026-06-16T03:25:51+00:00`  
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- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:25:02.674Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: ****What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?**** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best ***recent*** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the prior best model by 14.1 absolute percentage points, establishing per-moment classification + boundary regression as the state-of-the-art TAL design at the time."

****Source:**** https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29 (primary)

****Supporting quote:**** "ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:25:06.098Z)

I'll verify this claim about ActionFormer's THUMOS14 performance. Let me search for the primary source and corroborating evidence.

3. assistant (2026-06-16T03:25:10.140Z)

Let me search for the primary source on ActionFormer's THUMOS14 results.

4. user (2026-06-16T03:25:17.179Z)

Web search results for query: "ActionFormer "ECCV 2022" THUMOS14 average mAP 66.8 71.0% tIoU 0.5 results"

Links: [{"title": "GitHub - BharathChalla/ActionFormer: ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/BharathChalla/ActionFormer"}, {"title": "GitHub - yafeng19/actionformer: Code release for ActionFormer (ECCV 2022)", "url": "https://github.com/yafeng19/actionformer"}, {"title": "GitHub - chrisindris/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/chrisindris/actionformer_release"}, {"title": "GitHub - happyharrycn/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/happyharrycn/actionformer_release"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers | Computer Vision ? ECCV 2022", "url": "https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29"}, {"title": "actionformer_release/README.md at main · happyharrycn/actionformer_release · GitHub", "url": "https://github.com/happyharrycn/actionformer_release/blob/main/README.md"}, {"title": "ECVA | European Computer Vision Association", "url": "https://www.ecva.net/papers/eccv_2022/papers_ECCV/html/7278_ECCV_2022_paper.php"}, {"title": "GitHub - csh415/actionformer_performance", "url": "https://github.com/csh415/actionformer_performance"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/pdf/2202.07925"}, {"title": "A Multi-Modal Transformer Network for Action Detection", "url": "https://arxiv.org/pdf/2305.19624"}]

Based on the search results, I found the information you're looking for:

ActionFormer achieves an average mAP of 66.8% (at tIoU thresholds [0.3:0.1:0.7]), with an mAP of 71.0% at tIoU=0.5 on the THUMOS14 dataset. The model outperforms the best prior model by 14.1 absolute percentage points and crosses the 60% mAP for the first time.

The paper was accepted to ECCV 2022, and these results represent significant improvements in temporal action localization performance.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-16T03:25:20.276Z)

Web search results for query: "ActionFormer THUMOS14 71.0 mAP tIoU 0.5 outperforming best prior model 14.1 points"

Links: [{"title": "ActionFormer: Localizing Moments of Actions with Transformers | Springer Nature Link", "url": "https://link.springer.com/content/pdf/10.1007/978-3-031-19772-7_29"}, {"title": "[2202.07925] ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/abs/2202.07925"}, {"title": "GitHub - BharathChalla/ActionFormer: ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/BharathChalla/ActionFormer"}, {"title": "GitHub - yafeng19/actionformer: Code release for ActionFormer (ECCV 2022)", "url": "https://github.com/yafeng19/actionformer"}, {"title": "GitHub - chrisindris/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/chrisindris/actionformer_release"}, {"title": "GitHub - happyharrycn/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/happyharrycn/actionformer_release"}, {"title": "actionformer_release/README.md at main · happyharrycn/actionformer_release · GitHub", "url": "https://github.com/happyharrycn/actionformer_release/blob/main/README.md"}, {"title": "GitHub - csh415/actionformer_performance", "url": "https://github.com/csh415/actionformer_performance"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/pdf/2202.07925"}, {"title": "(PDF) ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.researchgate.net/publication/358655346_ActionFormer_Localizing_Moments_of_Actions_with_Transformers"}]

Based on the search results, I found the information you were looking for:

ActionFormer achieves 71.0% mAP at tIoU = 0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points. ActionFormer is a model designed to identify actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows. ActionFormer combines a multiscale feature representation with local self-attention, and uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries.

ActionFormer also demonstrates strong results on ActivityNet 1.3 (36.6% average mAP) and EPIC-Kitchens 100 (+13.5% average mAP over prior works).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:25:22.898Z)

The search results corroborate the claim. Let me verify directly against the primary source (arXiv abstract) to confirm the exact wording, since this matters for the "establishing per-moment classification + boundary regression as SOTA" part of the claim.

7. user (2026-06-16T03:25:30.917Z)

ActionFormer Performance Metrics

Based on the abstract, here are the exact reported numbers:

THUMOS14 Results:

The paper states: "ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points."

ActivityNet 1.3 Results:

"Further, ActionFormer demonstrates strong results on ActivityNet 1.3 (36.6% average mAP)"

Method Description:

The approach is characterized as identifying "actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows." The model employs "a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries."

The key innovation combines "a multiscale feature representation with local self-attention" to achieve these improvements over prior work.

8. assistant (2026-06-16T03:25:42.037Z)

The primary source (arXiv abstract) confirms the claim verbatim. Let me assess all parts of the claim against the evidence.

Verification analysis:

1. **"71.0% mAP at tIoU=0.5 on THUMOS14"** ? Confirmed verbatim in the arXiv abstract and the ECCV 2022 paper.
2. **"outperforming the prior best model by 14.1 absolute percentage points"** ? Confirmed verbatim in the abstract ("outperforming the best prior model by 14.1 absolute percentage points").
3. **"per-moment classification + boundary regression"** ? Confirmed. The abstract describes "a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries" ? this is exactly per-moment (anchor-free) classification plus boundary estimation/regression.
4. **"state-of-the-art TAL design at the time"** ? Reasonable and well-supported. The paper (ECCV 2022) reports crossing 60% average mAP "for the first time" and the +14.1 point gap at tIoU=0.5 substantiates SOTA-at-the-time. This is a defensible characterization, not an

overreach, for 2022. (Newer methods like TriDet later exceeded it, but the claim is explicitly scoped "at the time.")

****Source quality:**** Primary source is the peer-reviewed ECCV 2022 paper (Springer/ACM DL) and its arXiv version (2202.07925) ? strong, appropriate for the claim's strength.

****Adversarial checks:**** I attempted to find contradicting evidence. No credible source disputes these figures; multiple independent mirrors (GitHub release READMEs, Springer, ECVA) repeat them consistently. The numbers are not cherry-picked benchmark spin ? they are the headline THUMOS14 metric. The one minor nuance is the word "establishing... as the state-of-the-art design," which is an interpretive framing, but it is properly time-scoped ("at the time") and well-supported by the paper's own claim of crossing 60% avg mAP for the first time.

The claim is well-supported, current (correctly scoped to 2022), and backed by a primary peer-reviewed source.

9. user (2026-06-16T03:25:47.616Z)

Structured output provided successfully

10. assistant (2026-06-16T03:25:51.727Z)

Verification complete. The claim is ****not refuted**** ? it is well-supported by the primary peer-reviewed source.

The arXiv abstract for ActionFormer (2202.07925, ECCV 2022) confirms verbatim both quantitative claims (71.0% mAP at tIoU=0.5 on THUMOS14; +14.1 absolute points over prior best) and the methodological characterization (anchor-free per-moment classification + boundary estimation/regression). The "SOTA at the time" framing is interpretive but properly time-scoped and corroborated by the paper's own "first to cross 60% average mAP" claim. Source quality (peer-reviewed venue + arXiv, consistent across independent mirrors) matches the claim's strength. Confidence: high.